Day 24 | Thursday

December 10, 2026

24

Skill clinics: practise your priority topic

9:00 AM–10:15 AM • Math • 75 min

Mixed-topic calculation practice

10:30 AM–12:00 PM • Calculation Practice • 90 min

Week 7 • 2 scheduled blocks • 165 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

9:00 AM–10:15 AM • 75 min

Skill clinics: practise your priority topic

9:00 AM–9:05 AM	**Start**	5
9:05 AM–9:10 AM	**Recall**	5
9:10 AM–9:25 AM	**Teach**	15
9:25 AM–9:40 AM	**Together**	15
9:40 AM–10:00 AM	**Practise**	20
10:00 AM–10:10 AM	**Check**	10
10:10 AM–10:15 AM	**Wrap up**	5

Math • Math Week 7, Lesson 3

10:30 AM–12:00 PM • 90 min

Mixed-topic calculation practice

10:30 AM–10:35 AM	**Start**	5
10:35 AM–10:40 AM	**Recall**	5
10:40 AM–10:55 AM	**Teach**	15
10:55 AM–11:10 AM	**Together**	15
11:10 AM–11:45 AM	**Practise**	35
11:45 AM–11:55 AM	**Check**	10
11:55 AM–12:00 PM	**Wrap up**	5

Calculation Practice • Math Week 7, Lesson 4

Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time

Skill clinics: practise your priority topic

Today’s goal: Repair a documented error and explain a corrected method independently.

Identify one target skill and the checking habit to use today.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Path A: 4A Q14–17 p.26; 5A Q12–15 p.33. B: 7A Q11–14 pp.45–46; 8C Q1–2 p.55. C: 12C Q1–2 p.79; 13A Q6,10 p.82.

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Recall and reconnect

Use each learner’s Week 6 exits and capstone work to choose one pathway. Do not assume all learners need trig extension.

Review previous feedback and attempt the retrieval prompt without displayed solutions.

Resources: Path A: 4A Q14–17 p.26; 5A Q12–15 p.33. B: 7A Q11–14 pp.45–46; 8C Q1–2 p.55. C: 12C Q1–2 p.79; 13A Q6,10 p.82.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Choose the pathway your work shows you need

1 Use Week 6 exits and the applied drawing work to identify the error you need to repair.

2 Path A uses fractions and decimals: 4A Q14–17 and 5A Q12–15. Explain regrouping, conversion or the tape-reading error in your selected work.

3 Path B uses conversions and formulas: 7A Q11–14 and 8C Q1–2. Show the conversion direction or missing-dimension method.

4 Path C uses right triangles and trig: 12C Q1–2 and 13A Q6,10. Use this pathway when the evidence and taught content call for it; not everyone needs trig extension.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

A correction needs an explanation and a retry

1. Keep the original working so the point where the method failed remains visible.
2. Beside the corrected method, explain what changed and how your chosen check detects the original error.
3. Partners explain the first two selected questions; continue the scheduled four-question practice in your chosen pathway.
4. Keep tool answers hidden until after the written attempt. Use the scheduled independent pathway check to show that you can avoid the error yourself.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Choose one pathway and explain the method

A • FRACTIONS AND DECIMALS
4A Q14–17 p.26; 5A Q12–15 p.33.
Tools: Numeracy Hub Calculator and tape-reading practice.

B • CONVERSIONS AND FORMULAS
7A Q11–14 pp.45–46; 8C Q1–2 p.55.
Tools: Unit Conversions and a Shape Canvas rectangle.

C • RIGHT TRIANGLES AND TRIG
12C Q1–2 p.79; 13A Q6,10 p.82.
Tools: Angles and Trig Module 3 and a Shape Canvas right triangle.

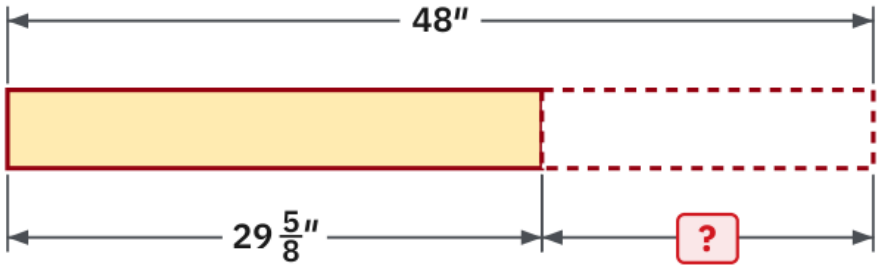
Instructor models one common pathway error. Partners solve the first two selected questions and explain the correction.

Exercise Book practice pathways • Tool answers stay hidden until the paper attempt

Textbook | page 66

Example 4.10 What is left of the bar?

A piece $29\frac{5}{8}$ in long is cut from a 48 in bar. Ignore the saw cut. How long is the offcut?



SOLUTION

1. There is no fraction in 48 to take $\frac{5}{8}$ from, so borrow 1: $48 = 47\ \frac{8}{8}$.
2. Fractions: $\frac{8}{8} - \frac{5}{8} = \frac{3}{8}$.
3. Whole numbers: $47 - 29 = 18$.

Path A only: regroup a whole number before subtracting fractional lengths.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Textbook | page 127

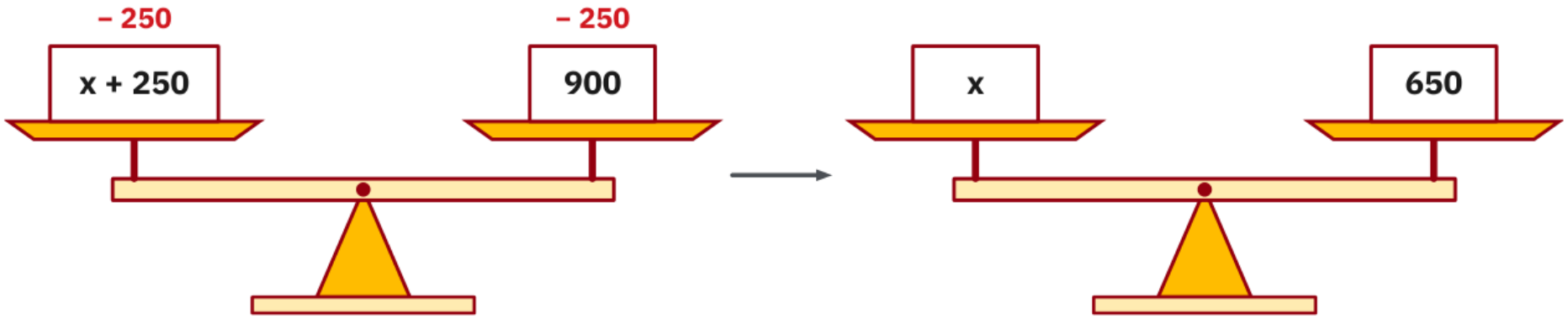


Figure 8.1 Take 250 off both pans and the scale stays level. The unknown x is left on its own.

Path B only: the same inverse operation keeps both sides equal.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Textbook | page 192

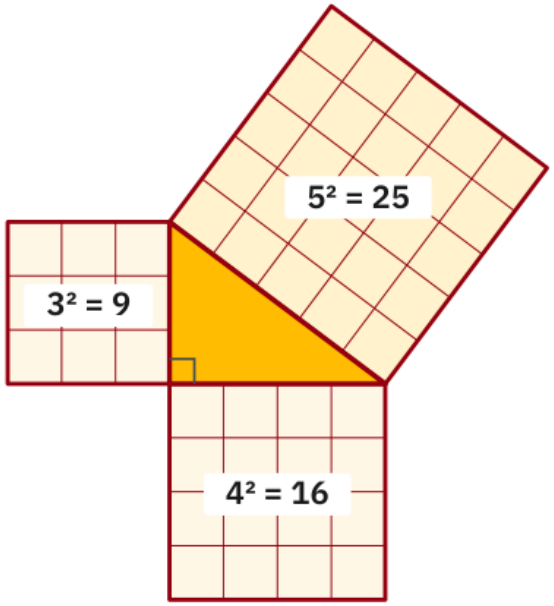


Figure 12.5 The squares on the sides of a 3-4-5 right triangle: 9 + 16 = 25.

Path C only: revisit the right-triangle relationship before solving.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Spot the mistake. Explain the correction.

Your corrected page now matches the tool answer, but the original error has been erased. Does that show which part of your method improved?

Discuss the method before checking the answer.

Retain before-and-after working, identify the incorrect step and explain the repair. Matching a displayed answer does not replace the scheduled independent pathway check.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Work together

Partners work the first two selected pathway questions and explain the correction in their own words.

Explain the method and check directions; work with a partner only where permitted.

Resources: Path A: 4A Q14–17 p.26; 5A Q12–15 p.33. B: 7A Q11–14 pp.45–46; 8C Q1–2 p.55. C: 12C Q1–2 p.79; 13A Q6,10 p.82.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Your scheduled practice

Complete four questions in the selected pathway, then one from another pathway if ready. Keep a before/after solution showing what changed; tool answers remain hidden until the attempt.

Keep numbered written solutions and units. Use tools only under the conditions stated for this block.

Resources: Path A: 4A Q14–17 p.26; 5A Q12–15 p.33. B: 7A Q11–14 pp.45–46; 8C Q1–2 p.55. C: 12C Q1–2 p.79; 13A Q6,10 p.82.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Open Numeracy Hub

ESSENTIAL SKILLS FOR ABORIGINAL FUTURES · SKILLS FOR SUCCESS: NUMERACY

ACCESS Trades Numeracy Hub

Every worked example shows the calculator keys. Open the calculator any time with the button at the bottom right.

Calculator TI-30Xa

1 Shape Studio | 2 Order of Operations | 3 Volume and Layout | 4 Angles and Trig | 5 Fabrication | 6 Pipe Trades | 7 Unit Conversions | 8 Mechanical Aptitude | 9 Coordinate Systems | 10 Measuring Tools | 11 Exam builder

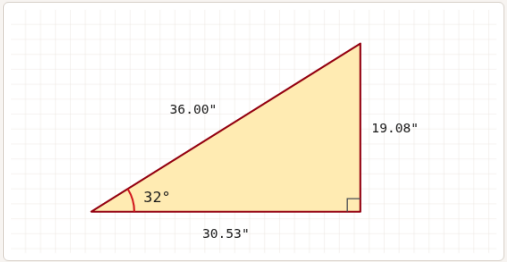
A– 100% A+ Hide operations

MODULE 3

Right-triangle trigonometry

Braces, gussets, ramps, cut angles and tapers are all right triangles. Trig finds the missing side or angle without laying it out full size.

SOH: sin = opposite ÷ hypotenuse. CAH: cos = adjacent ÷ hypotenuse. TOA: tan = opposite ÷ adjacent. Set your calculator to degrees (DEG).



Worked example

A brace 36.00" long leans at 32° from the floor. How high up the post does it reach, and how far out along the floor?

1. Label the sides from the 32° angle. The brace is the hypotenuse. The rise is opposite. The run is adjacent.
2. Rise uses SOH: opposite = hypotenuse × sin 32° = 36.00 × 0.5299 = 19.08".
3. Run uses CAH: adjacent = hypotenuse × cos 32° = 36.00 × 0.8480 = 30.53".
4. Check with Pythagoras: $19.08^2 + 30.53^2 \approx 36.00^2$.

Answer: Rise (opposite) 19.08 in, Run (adjacent) 30.53 in

TRY OTHER NUMBERS

Angle 32°

Brace length (hypotenuse) 36.00"

On the TI-30Xa Start with ON/C ON/C and DEG on the screen.

Rise

36 × 32 SIN =

SCREEN 19.07709351 Run on calculator

Calculator

Choose only your assigned pathway. Trigonometry is for Path C.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Practise in Numeracy Hub and Shape Canvas

1. Choose the tool for your selected pathway only: A—Numeracy Hub Calculator and tape practice; B—Unit Conversions and a Shape Canvas rectangle; C—Angles and Trig Module 3 and a Shape Canvas right triangle.

2. Complete the selected source-book work on paper first, with units and a visible method. Leave tool answers hidden during the attempt.

3. Use the pathway tool to check the attempt and explain the correction beside the original working. Retain that record for the scheduled independent pathway check.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Practise from the Exercise Book | 4A Q14a

Add. (a) $\frac{5}{16} + \frac{3}{4}$

Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Practise from the Exercise Book | 5A Q12

Change $\frac{7}{8}$ and $3\frac{1}{4}$ to decimals.

Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Practise from the Exercise Book | 7A Q12

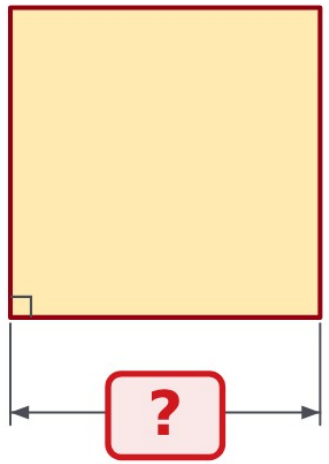
Convert 2 1/4 in to mm.

Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Practise from the Exercise Book | 8C Q2

The square has an area of 3,136 mm². Find the side.

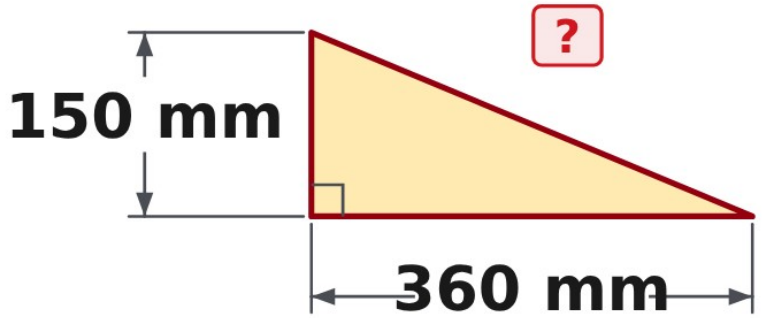


Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Practise from the Exercise Book | 12C Q1

Find the hypotenuse.



Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Practise from the Exercise Book | 13A Q6

Adjacent 900 mm, angle 24°. Find the opposite side to two decimal places.

Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Check independently

Use Path A: 4A Q14(b); B: 7A Q11; or C: 13A Q10. Explain the error avoided.

Submit work and demonstrate the checking strategy; identify a remaining difficulty.

Resources: Path A: 4A Q14–17 p.26; 5A Q12–15 p.33. B: 7A Q11–14 pp.45–46; 8C Q1–2 p.55. C: 12C Q1–2 p.79; 13A Q6,10 p.82.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Independent check | 4A Q14b

Add. (b) $7\frac{3}{8} + 2\frac{15}{16}$

Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Independent check | 7A Q11

Convert 5/8 in to mm.

Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Independent check | 13A Q10

Opposite 350 mm, adjacent 800 mm. Find the angle to two decimal places.

Use only the question for your selected skill pathway.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Wrap up and choose the next step

Identify what improved, what needs another attempt and which resource will support the next step.

Name the math and another essential skill used; record a concrete follow-up action.

Resources: Path A: 4A Q14–17 p.26; 5A Q12–15 p.33. B: 7A Q11–14 pp.45–46; 8C Q1–2 p.55. C: 12C Q1–2 p.79; 13A Q6,10 p.82.

Textbook Chapter 4–5 pp.56–87; Chapter 7–8 pp.108–134; Chapter 12–13 pp.188–209, selected

Break

10:15 AM–10:30 AM

Return ready for: Mixed-topic calculation practice

This break is excluded from teaching minutes.

Mixed-topic calculation practice

Today’s goal: Apply taught methods to fresh practice, retain working and use feedback to choose a revision priority.

Identify one target skill and the checking habit to use today.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback: 3A Q8–9 p.19; 6B Q2 p.41; 7A Q11 p.45; 8C Q1–2 p.55; 10A Q5–6 p.63; 12C Q1–2 p.79.

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Recall and reconnect

Try to choose one personal error to watch for and identify their calculator model.

Review previous feedback and attempt the retrieval prompt without displayed solutions.

Resources: Paper fallback: 3A Q8–9 p.19; 6B Q2 p.41; 7A Q11 p.45; 8C Q1–2 p.55; 10A Q5–6 p.63; 12C Q1–2 p.79.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Make the generated set reproducible

1. Start with All off and enable only taught order, conversion, area/perimeter, volume or right-triangle topics.
2. Use Easy, 10 questions, diagrams on and calculator keys on; keep worked answers off during the attempt.
3. Retain the exam code before use. The instructor previews the set and removes untaught or trade-specific material.
4. Use the previewed ten-question set or the exact ten paper fallback items. This is formative practice, not an institutional pass/fail test.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Record the method before checking the result

1. Start each response with the question number or saved code so the working can be matched to its prompt.

2. Write the formula or operation, then working, units and the requested final rounding.

3. Include an estimate or second-method check; a calculator display alone does not explain your method.

4. After completion, check and correct. The scheduled independent retry uses new numbers for one missed type from the same taught-topic settings.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Keep a reproducible ten-question record

Demonstrate reading a question, saving its code and using requested rounding.
Work one separate example together; keep practice-set answers hidden.

PAPER FALLBACK • ITEMS 1–5

1. 3A Q8 — p.19
2. 3A Q9 — p.19
3. 6B Q2 — p.41
4. 7A Q11 — p.45
5. 8C Q1 — p.55

PAPER FALLBACK • ITEMS 6–10

6. 8C Q2 — p.55
7. 10A Q5 — p.63
8. 10A Q6 — p.63
9. 12C Q1 — p.79
10. 12C Q2 — p.79

Response format • Question number/code → formula or operation → working → units and final rounding → estimate or second-method check.

Exercise Book required for full numbered questions • Use the assigned paper questions if needed

Textbook | page 219

Order of operations

Brackets, Exponents, Division and Multiplication (left to right), Addition and Subtraction (left to right)	**BEDMAS**

Perimeter and distance around

Rectangle	$P = 2 \times (l + w)$
Any shape with straight sides	**P = sum of the sides**
Circle	$C = \pi \times D$
Diameter and radius	$D = 2 \times r$

Use only formulas for the topics selected in today's practice set.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Textbook | page 219

Right triangles

Pythagorean theorem	$c^2 = a^2 + b^2$
Angles of a triangle	$A + B + C = 180°$
Sine	$\sin \theta$ = opposite ÷ hypotenuse
Cosine	$\cos \theta$ = adjacent ÷ hypotenuse
Tangent	$\tan \theta$ = opposite ÷ adjacent

Right-triangle references apply only when those taught topics are selected.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Spot the mistake. Explain the correction.

You reveal worked answers while you are still completing the set, then copy the method. Can the record show which method you chose independently?

Discuss the method before checking the answer.

Keep worked answers hidden until the whole attempt is complete. Then use Check my answers, reveal worked answers for corrections, and retain the code/results so the attempt and correction can be distinguished.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Work together

Work one separate demonstration question; state the formula and units before keys. Keep assessment/practice answers hidden.

Explain the method and check directions; work with a partner only where permitted.

Resources: Paper fallback: 3A Q8–9 p.19; 6B Q2 p.41; 7A Q11 p.45; 8C Q1–2 p.55; 10A Q5–6 p.63; 12C Q1–2 p.79.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Your scheduled practice

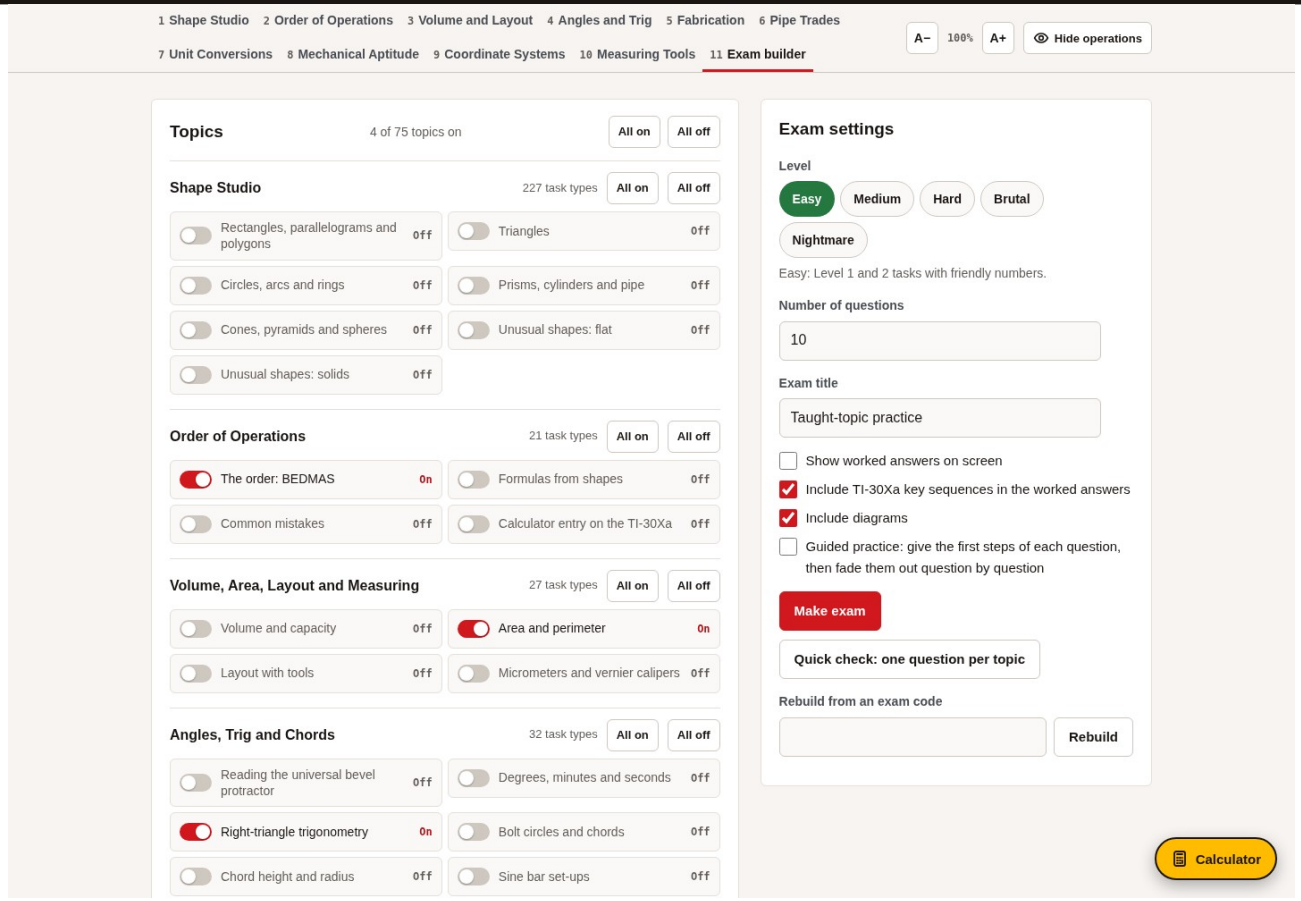
Complete the previewed 10-question Numeracy Hub set, or the ten paper fallback questions. Keep written methods. Use Check my answers only after completion; reveal worked answers for correction and save the code/results manually.

Keep numbered written solutions and units. Use tools only under the conditions stated for this block.

Resources: Paper fallback: 3A Q8–9 p.19; 6B Q2 p.41; 7A Q11 p.45; 8C Q1–2 p.55; 10A Q5–6 p.63; 12C Q1–2 p.79.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Practise in Numeracy Hub

1. Configure Numeracy Hub Exam builder: All off; only the scheduled taught topics; Easy; 10 questions; diagrams and calculator keys on; worked answers off. Save the code and use the instructor-previewed set.

2. Complete the ten questions with numbered written methods and units. Use the exact scheduled paper fallback if the generated set is not used.

3. After completion, use Check my answers and then worked answers for corrections; save code/results manually. For the scheduled check, explain one correction and independently retry one missed type with new numbers from the same settings.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 1 of 10 | 3A Q8

$6^2 - 4 \times 5$

Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 2 of 10 | 3A Q9

$$(6 - 4)^2 \times 5$$

Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 3 of 10 | 6B Q2

A fabrication job needs 18 sheets of 4 ft × 8 ft plate. Add a 15% allowance and round up to whole sheets.

Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 4 of 10 | 7A Q11

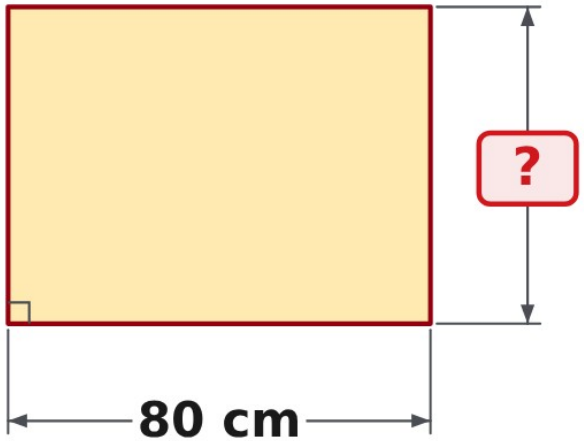
Convert 5/8 in to mm.

Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 5 of 10 | 8C Q1

The rectangle has an area of 4,800 cm². Find the width.

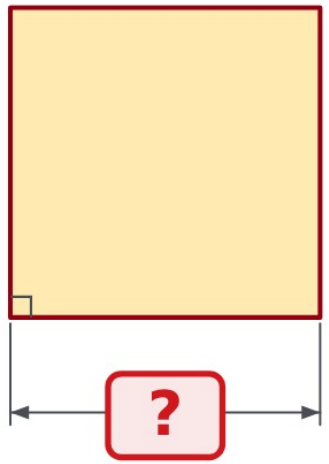


Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 6 of 10 | 8C Q2

The square has an area of 3,136 mm². Find the side.



Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 7 of 10 | 10A Q5

Find the area of a rectangle 840 mm by 350 mm.

Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 8 of 10 | 10A Q6

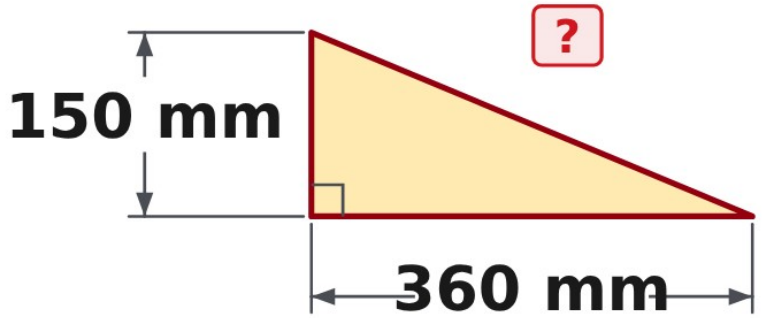
Find the area of a square with 1.6 m sides.

Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 9 of 10 | 12C Q1

Find the hypotenuse.

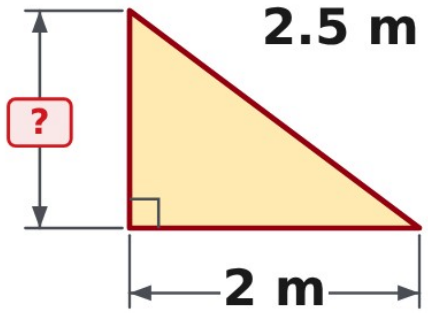


Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Paper fallback 10 of 10 | 12C Q2

Find the missing leg.



Write the operation, working and units before checking.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Check independently

Explain one corrected item and independently redo one missed type using new numbers from the same tool settings.

Submit work and demonstrate the checking strategy; identify a remaining difficulty.

Resources: Paper fallback: 3A Q8–9 p.19; 6B Q2 p.41; 7A Q11 p.45; 8C Q1–2 p.55; 10A Q5–6 p.63; 12C Q1–2 p.79.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed

Wrap up and choose the next step

Identify what improved, what needs another attempt and which resource will support the next step.

Name the math and another essential skill used; record a concrete follow-up action.

Resources: Paper fallback: 3A Q8–9 p.19; 6B Q2 p.41; 7A Q11 p.45; 8C Q1–2 p.55; 10A Q5–6 p.63; 12C Q1–2 p.79.

Textbook formula sheet p.219; conversion factors pp.220–221; relevant chapters as needed